A Comprehensive Survey on Source-Free Domain Adaptation

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(Survey Paper)

***Abstract*—Over the past decade, domain adaptation has become a widely studied branch of transfer learning which aims to improve performance on target domains by leveraging knowledge from the source domain. Conventional domain adaptation methods often assume access to both source and target domain data simultaneously, which may not be feasible in real-world scenarios due to privacy and confidentiality concerns. As a result, the research of Source-Free Domain Adaptation (SFDA) has drawn growing attention in recent years, which only utilizes the source-trained model and unlabeled target data to adapt to the target domain. Despite the rapid explosion of SFDA work, there has been no timely and comprehensive survey in the field. To fill this gap, we provide a comprehensive survey of recent advances in SFDA and organize them into a unified categorization scheme based on the framework of transfer learning. Instead of presenting each approach independently, we modularize several components of each method to more clearly illustrate their relationships and mechanisms in light of the composite properties of each method. Furthermore, we compare the results of more than 30 representative SFDA methods on three popular classification benchmarks, namely Office-31, Office-home, and VisDA, to explore the effectiveness of various technical routes and the combination effects among them. Additionally, we briefly introduce the applications of SFDA and related fields. Drawing on our analysis of the challenges confronting SFDA, we offer some insights into future research directions and potential settings.**

***Index Terms*—Computer vision, data-free learning, domain adaptation, transfer learning.**

I. Introduction

Deep neural networks have demonstrated impressive performance in supervised learning tasks due to their exceptional ability to generalize. However, collecting sufficient training data can be a costly and challenging task, primarily due to factors such as financial constraints and privacy concerns. For example, the manual annotation of a single cityscape image for semantic segmentation can take up to 90 minutes. To alleviate this issue, transfer learning has been proposed as a viable solution to enable cross-domain knowledge transfer under label-deficient conditions [1], [2].

Unsupervised Domain adaptation (UDA) [3], [4], [5], as an important branch of transfer learning, focuses on how to improve model performance on the unlabeled target domain with the assistance of labeled source domain data. In UDA, it is generally assumed that the access to both source and target domain data is available throughout the adaptation process. However, in numerous practical applications, accessing source domain data may not be feasible due to concerns regarding personal privacy, confidentiality, and copyright. Furthermore, storing the entire source dataset on devices with limited resources for training purposes can be resource-intensive. To mitigate the reliance of domain adaptation methods on source data, Source-Free Domain Adaptation (SFDA) [6], [7] has been introduced in recent years, which has gained significant attention and been extensively studied in various fields such as image classification [8], [9], semantic segmentation [10], [11] and object detection [12], [13].

The principal distinction between SFDA and UDA lies in the treatment of the source and target domain data during the training phase. As shown in Fig. 1, UDA typically employs a single-stage training process where both supervised learning on the source domain data and distribution alignment between the source and target domains are performed concurrently. In contrast, SFDA follows a two-stage training process, where supervised training is initially conducted using the source domain data, followed by separate training using only the target domain data. This two-stage process effectively achieves data isolation between the domains. The mainstream methods of UDA can be roughly categorized into three types: (a) aligning the source and target domain distributions by designing specific metrics [14], [15], [16]. (b) learning domain-invariant features through adversarial learning [17], [18], [19]. (c) fine-tuning the model in a self-supervised manner [20], [21], [22]. Compared to UDA methods, SFDA approaches face two key challenges. First, it becomes impractical to directly align the distributions of the source and target domains since the source domain data is unavailable. This

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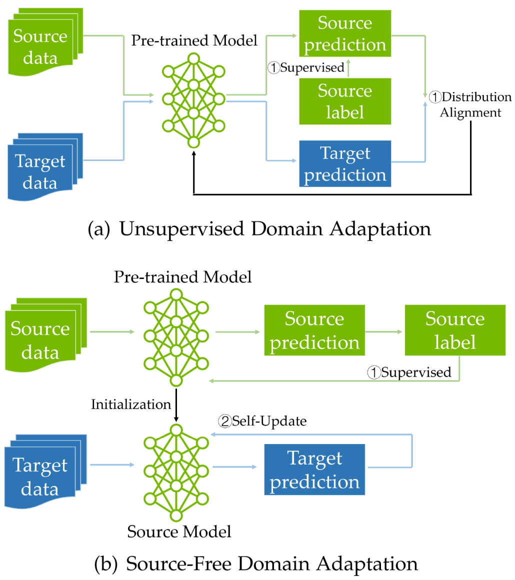

Fig. 1. Comparison of UDA and SFDA settings.

poses a major obstacle to applying traditional UDA methods in the SFDA context. Second, the absence of labeled source supervision during training can result in unknown model deviation and lead to a degradation of accuracy. These challenges highlight the need for specialized techniques and approaches in SFDA. Despite the considerable efforts dedicated to SFDA, there is currently no comprehensive survey that encompasses all the works and provides a summary of the recent progress in the field. To address this gap, our objective is to present a timely and comprehensive review of the recent advances in SFDA. In terms of literature selection, we comprehensively considered the citation frequency, authority of the publishing journal or conference, as well as the timeliness of the literature, in order to ensure that our survey accurately reflects the latest developments and diversity in the field.

Additionally, our investigation indicates that the majority of SFDA methods are composed of multiple technical components, with each component potentially corresponding to a category within our taxonomy. Therefore, unlike previous surveys on domain adaptation which treated each work independently, our aim is to modularize each SFDA method and categorize them accordingly, in order to reveal the interrelatedness among different research directions. The goal of this survey is to discuss the problem formulation of SFDA, summarize the current advances in this field, and highlight the distinctive components of SFDA techniques to provide a comprehensive understanding of this area and inspire more ideas and applications. The main contributions of this survey can be summarized as follows.

- We propose an overall taxonomy framework for the newly emerged SFDA setting. This survey fills a gap in the existing literature by providing a timely and comprehensive summary of the latest SFDA methods and applications.
- We modularize more than 30 representative works and compare the results on the most widely used classification datasets, i.e., Office-31, Office-home, and VisDA. The combination of different components is visually presented and comparatively analyzed, which may be instructive for further research in the SFDA setting.

The remainder of this survey is structured as follows. In Section II, we discuss various research topics related to SFDA. Section III provides notations, preliminary knowledge in SFDA, and a brief description of our taxonomy. Sections IV and V present a comprehensive review of data-centric and model-centric SFDA methods, respectively. In Section VI, we compare existing SFDA methods on three mainstream classification datasets. Section VII discusses the potential applications of SFDA, while Section VIII presents visions for the future directions of SFDA. Finally, we conclude the paper in Section IX.

II. Related Research Topics

In this section, we discuss several research topics related to source-free domain adaptation, highlighting their interconnections and distinctions. These topics are systematically presented and compared in Table I, with the detailed information provided subsequently.

A. *Data-Free Knowledge Distillation*

Knowledge distillation is proposed by Hinton et al. [23] to transfer knowledge from a large-scale teacher network to a simplified student network, which has been used in model compression [24], [25], domain adaptation [26], [27] and multi-modal learning [28], [29]. Recently, Data-Free Knowledge Distillation (DFKD) has gained significant attention due to privacy concerns. DFKD typically involves generating a training dataset and then training a compact student model using the generated dataset. Various methods have been proposed to generate training images, including feature statistics alignment [30], [31], teacher confidence enhancement [32], [33] and adversarial generation [34], [35]. DFKD focuses on knowledge transfer within the same domain, while SFDA aims to transfer knowledge between domains. In some SFDA methods, knowledge distillation techniques are utilized to preserve the knowledge from the source domain [36], [37], which mostly use knowledge distillation to preserve the knowledge of the source domain. As a potential extension to SFDA, DFKD can be employed to generate virtual source domain data, helping to alleviate the unavailability problem of source data.

B. *Domain Adaptation and Privacy-Concerned DA*

Domain Adaptation (DA) is a representative problem studied in transfer learning. Among different variants, Unsupervised Domain Adaptation (UDA) is a widely studied subproblem where a labeled source domain is utilized to enhance the model's performance on the unlabeled target domain under the domain shift. UDA methods can be concluded into three categories. The first one is metric learning [14], [38], which mainly defines a certain criterion that measures the distribution distance between two domains, and explicitly minimizes this criterion to achieve inter-domain alignment. The most representative method is DAN [14], which employs maximum mean discrepancy (MMD) [39] in a reproducing kernel Hilbert space to explicitly align different domain distributions. The second category is adversarial learning, which is based on the idea of

TABLE I
SFDA VERSUS OTHER RELATED SETTINGS

	No source data	Source model	Unlabeled target	Extra source information	No training
Data-free KD	✔	-	-	-	✗
Domain Generalization	✗	✔	-	-	✗
Few-shot Learning	✗	✔	✔	-	✗
Federated DA	✗	✔	✔	✗	✗
Zero-shot DA	✗	✔	✗	✗	✗
Test-time DA	✔	✔	✔	✗	✔
Source-relaxed DA	✔	✔	✔	✔	✗
Source-free DA	✔	✔	✔	✗	✗

GAN [40]. The most typical way for adversarial learning is to employ an extra domain discriminator and conduct a mini-max game between it and the feature extractor [18], [41]. Another implementation of adversarial learning is to employ multiple task-specific classifiers and make an intra-network mini-max game between the feature extractor and these classifiers [42], [43]. The final category is self-supervised training [44], [45]. It typically involves self-supervision by pseudo-labeling the target domain and has gained significant popularity in the SFDA setting. Federated DA [46], [47] aims to transfer knowledge from decentralized private nodes to a new node with a distinct data domain. Zero-shot DA [48], [49] strives to acquire task-specific knowledge from the source domain when the target domain is inaccessible during training. Test-time DA [50], [51] can be considered as online SFDA, wherein the model adapts directly to the target domain test data. Source Relaxed DA (SRDA) [52], [53] is quite similar to the SFDA setting, but it requires additional source information along with a well-trained source model.

C. Related Generalization Tasks in Transfer Learning

In the context of transfer learning, domain generalization (DG) [54] and few-shot learning tasks [55], [56], [57] bear a certain degree of similarity to SFDA, but they require differentiation. Specifically, the emphasis of the domain generalization task lies in generalizing the model from the source domain to unseen domains. It is noteworthy that when considering an unknown domain as the target domain, the performance of the recent domain generalization approach [58] even surpasses that of several UDA methods. However, in situations where the source data is not available due to privacy security or storage efficiency considerations, domain generalization methods are not applicable. Few-shot learning [59], [60], [61], [62] involves training a model with a small number of labeled samples in one domain, and then making predictions or generalizations in another related domain including unseen classes. In contrast to domain adaptation, few-shot learning mainly focuses on predicting new categories under class inconsistency, rather than the inconsistency in domain distribution.

III. OVERVIEW

In this section, we introduce some notations and preliminary knowledge related to SFDA, with a brief description of several settings in SFDA methods. Finally, we present our taxonomy of SFDA and the underlying motivation.

TABLE II
NOTATIONS RELATED TO SFDA

Symbol	Description
n	Number of instances
$\mathcal{D}$	Domain
$\mathcal{X}$	Instance set
d	Feature dimensions
$\mathcal{F}(\mathcal{X})$	Feature vector
$\mathcal{Y}\|\mathcal{X}$	Prediction for instance $\mathcal{X}$
Φ	Domain space
$\mathbf{L}$	Source label set
$\mathcal{L}$	Loss function
δ	Soft-max operation
K	Number of classes
$\tilde{z}$	Noise vector
Θ	Model parameters
$\mathcal{F}$	Feature extractor
$\mathcal{C}$	Classifier
$\mathcal{D}$	Domain discriminator
G	Generator
$\mathcal{M}$	Model to be trained
$\mathbf{M}$	Module
$\mathcal{B}$	Memory bank
μ	Means
σ	Standard deviation
$\mathbb{E}$	Mathematical expectation
P	Marginal distribution
$\mathcal{Q}$	Query
$\mathcal{K}$	Key
$\mathcal{V}$	Value
$\mathcal{N}(\mu, \sigma^2)$	A Gaussian distribution with mean μ and variance σ^2

A. Notations and Definitions

To ensure consistency with prior literature, we have made efforts to utilize the same notations as employed in existing surveys on transfer learning [63] and deep domain adaptation [64], [65]. The notations used are summarized in Table II. Additionally, we present essential definitions that are closely linked to the SFDA problem.

Definition 1: (Domain in SFDA) A domain usually contains a data set Φ and a corresponding label set $\mathbf{L}$, where the data set consists of the instance set $\mathcal{X}$ drawn from the marginal distribution $\mathcal{P}(\mathcal{X})$ and corresponding dimension d. In SFDA, the source domain $\mathcal{D}^s = \{\{\mathcal{X}^s, \mathcal{P}(\mathcal{X}^s), d^s\}, \mathbf{L}^s\}$ is usually complete, but note that it is only available during pre-training. The label set of the target domain $\mathcal{D}^t = \{\{\mathcal{X}^t, \mathcal{P}(\mathcal{X}^t), d^t\}\}$ is not available and its marginal distribution is also different from the source domain. In typical SFDA, we assume a closed setting, i.e., the label spaces of the source and target domains are the same.

Definition 2: (Task of SFDA) Given a model $\mathcal{M}$ trained on the source domain $\mathcal{D}^s = \{\{\mathcal{X}^s, \mathcal{P}(\mathcal{X}^s), d^s\}, \mathbf{L}^s\}$, the process of SFDA consists of two stages. In the first stage (source training), the pre-trained model is trained by labeled source data under supervised learning. In the second stage (adaptation), the source

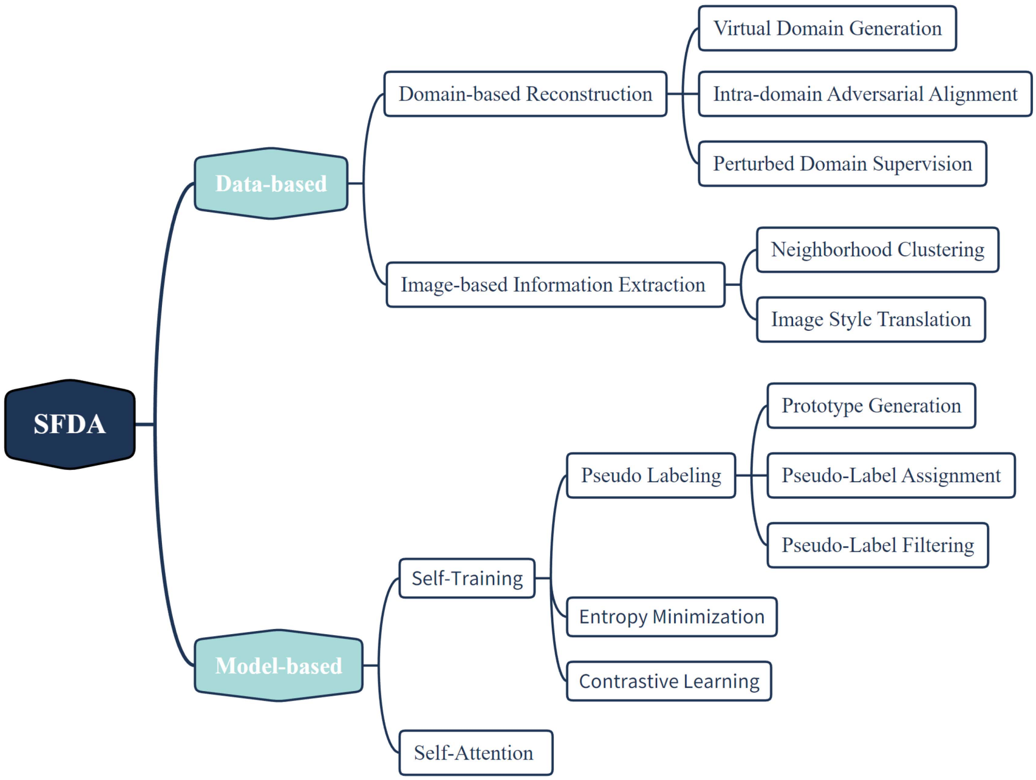

Fig. 2. Taxonomy of SFDA methods.

domain data is unavailable, and the goal of SFDA is to adapt the source-trained model to the target domain with unlabeled target data $\Phi^t = \{\{\mathcal{X}^t, \mathcal{P}(\mathcal{X})^t, d^t\}\}$, i.e., to get high accuracy for outputs $\mathcal{M}(\mathcal{X}^t) = P(\mathcal{Y}^t|\mathcal{X}^t)$, where $P(\mathcal{Y}^t|\mathcal{X}^t)$ is the prediction result of target data $\mathcal{X}^t$.

It is noteworthy that the aforementioned definitions may also facilitate certain extensions. Regarding the definition of the domain, SFDA may include multiple source domains, denoted as $\mathcal{D}^s = \sum_{i=1}^{m} \mathcal{D}_i^s$, where m represents the number of source domains. Additionally, it is also possible for open-set scenarios such as Universal Source-Free Domain Adaptation [8]. As for the definition of the task, it is feasible to annotate a small subset of target domain data to maximally boost the model performance, as observed in several extended SFDA settings (e.g., Active SFDA [66], [67]). The subsequent section will discuss various SFDA variants.

B. Overview of the Settings in SFDA

Generally, domain adaptation can be divided into homogeneous domain adaptation and heterogeneous domain adaptation according to whether the source and target domain feature space are identical. In heterogeneous domain adaptation [68], both the distribution space and feature space of the source and target may be different, i.e., $P(\mathcal{X}^s) \neq P(\mathcal{X}^t)$ and $d^s \neq d^t$, so both distribution adaptation and feature space adaptation may be required. While for homogeneous domain adaptation, the feature spaces of the source and target domains are identical, and only distribution adaptation is required. Among the existing SFDA methods, most of them only involve homogeneous domain adaptation, which is also the discussion focus of this survey.

According to the label-set correspondence between the source and target domains, the settings can be classified as closed-set, partial, and open-set, and the relationship between them is shown in Fig. 3. Based on the number of source and target domains, there are also single-source, multi-source, and multi-task settings. In this survey, we mainly focus on the most general closed-set single-source SFDA setting, and other settings are also briefly outlined.

C. Taxonomy

In this work, we propose to divide SFDA into two main directions: data-driven and model-driven. This division is inspired by a survey on transfer learning [63], which categorizes methods based on two perspectives: data-based and model-based. Moreover, through extensive literature review, we have discovered that the holistic SFDA research also well aligns with these two directions. Therefore, we have adopted this taxonomy for a systematic summary of the current SFDA research. Specifically, data-driven methods primarily focus on data reconstruction or the extraction of intrinsic feature information from existing data. In contrast, model-driven methods are centered around the source model and typically update it by drawing inspiration from self-supervised learning techniques. The overall taxonomy of SFDA methods is depicted in Fig. 2.

There are two main categories of data-driven methods: domain-based reconstruction and image-based information extraction. (a) Domain-based reconstruction aims to compensate for missing source domain data by reconstructing a domain or creating further divisions within the target domain, thereby enhancing the model's generalization ability by aligning the distribution between the two domains. Concretely, it can be subdivided into virtual domain generation, intra-domain adversarial alignment, and perturbed domain supervision. (b) The other approach is to extract intrinsic feature information from the target domain. Neighborhood clustering assumes the presence of clear clustering distribution and spatial structures within the target domain. Its objective is to elucidate the target domain's clustering by ensuring the consistency of neighboring nodes, thereby facilitating prediction. For better recognition by the source classifier, Image Style Translation attempts to convert the style information of target image into the source style, while maintaining the content information unchanged.

Model-driven approaches assume that the source model possesses a certain level of generalization over the target domain, owing to the relevance between the source and target domains. As a result, the model can be fine-tuned by exploring its outputs on the target data, which is a self-training scheme inherited from the concept of semi-supervised learning [69], [70], [71]. This self-training scheme can be further divided into three categories: pseudo-labeling, entropy minimization, and contrastive learning. Notably, we pay particular attention to the pseudo-labeling methods, as they are the most commonly used in SFDA. Pseudo-labeling is typically achieved by first pseudo-labeling the high-confidence samples in the target domain, and then optimizing the model using the obtained pseudo-labels. Therefore, obtaining high-quality pseudo-labels is the main concern of this category of methods. In this paper, we aim to comprehensively discuss this technical route based on each process, including prototype generation, pseudo-label assignment, and pseudo-label filtering. Furthermore, the self-attention mechanism is commonly employed as an auxiliary module, as accurately localizing the target positions within images holds significant importance in the domain adaptation task.

IV. Data-Based Methods

Within the scope of data-based approaches for SFDA, we can identify two distinct directions. The first direction focuses on tackling the problem of unavailable source domain data. In this scenario, the main purpose involves imitating the source domain data by utilizing the source domain information embedded in the model. We refer to this direction as Domain-based Reconstruction. The second direction addresses the challenge of unannotated target domain data. In this case, methods aim to explore the inherent data structure or clustering information hidden in the unlabeled target domain data, enabling the domain adaptation task to be accomplished solely using the target domain data. We refer to this direction as Image-based Information Extraction.

A. Domain-Based Reconstruction

The primary objective of this broad class method is to reconstruct a new domain for the supervision of the target domain. In general, there exist three research lines to constructing a new domain. The first approach involves generating a virtual source domain, while the second approach involves dividing the target domain into source-like and target-like parts according to prediction confidence, followed by performing an intra-domain adversarial to achieve alignment. The third approach involves constructing a perturbed target domain for robust learning. To facilitate comparison, we have depicted the three frameworks in Fig. 4.

1) Virtual Domain Generation: In unsupervised domain adaptation, a typical paradigm is to supervise the model on the source domain, and meanwhile employ some techniques to align the distributions of the source and target domains directly. The overall training process can be formulated as:

$$\mathcal{L}_{UDA_M} = \mathcal{L}_{cls}\left(\mathcal{C}\left(\mathcal{F}\left(\mathcal{X}^s\right)\right), \mathcal{Y}^s\right) + \mathcal{L}_{div}\left(\mathcal{F}\left(\mathcal{X}^s\right), \mathcal{F}\left(\mathcal{X}^t\right)\right), \tag{1}$$

where $\mathcal{L}_{cls}$ is the supervised loss in the source domain, usually using cross-entropy. $\mathcal{L}_{div}$ denotes the criterion used for alignment. In the scenario of SFDA, it is not feasible to access $\mathcal{X}^s$ during the adaptation phase. As a result, virtual domain generation methods strive to construct an alternative domain that can effectively substitute the role of the source domain. To expand the aforementioned equation, the following extension can be proposed:

$$\mathcal{L}_{vdg} = \mathcal{L}_{gen}\left(\mathcal{C}\left(\mathcal{F}\left(\mathcal{X}^v\right)\right)\right) + \mathcal{L}_{div}\left(\mathcal{F}\left(\mathcal{X}^v\right), \mathcal{F}\left(\mathcal{X}^t\right)\right), \tag{2}$$

where $\mathcal{X}^v$ denotes the virtual domain and $\mathcal{L}_{gen}$ denotes the generative loss of the virtual domain. Comparing to (1), SFDA should be concerned with the quality of virtual domain generation in addition to the alignment between domains. When the virtual domain is generated, the alignment problem of SDFA becomes analogous to that of UDA. For example, in VDM-DA [72], ADDA [18] is employed, and in STDA [73], MMD [39] is utilized as an alignment metric. Therefore, the primary focus of this section is to address the methodology for generating the virtual domain. Technically, the virtual domains can be generated in two ways: based on adversarial generation or based on Gaussian distribution. Drawing on the idea of GAN [40], the adversarial generation can be expressed as:

$$\mathcal{L}\left(\mathcal{C}, \mathcal{F}, \mathcal{D}, G\right) = \mathcal{L}_{adv} + \mathcal{L}_{con}, \tag{3}$$

where $\mathcal{L}_{con}$ is the consistency loss used to constrain the semantic consistency across domains. $\mathcal{L}_{adv}$ represents the adversarial loss between the generator and the domain discriminator, and is usually based on the following equation:

$$\begin{aligned} \mathcal{L}_{adv}(G) &= \mathbb{E}_{y,\widetilde{z}}\left[\log\left(1 - \mathcal{D}\left(G\left(y, \widetilde{z}\right)\right)\right)\right], \\ \mathcal{L}_{adv}\left(\mathcal{D}\right) &= \mathbb{E}_{x_t \sim \mathcal{X}_t}\left[\log \mathcal{D}\left(x_t\right)\right] + \mathbb{E}_{y,\widetilde{z}}\left[\log\left(1 - \mathcal{D}\left(G\left(y, \widetilde{z}\right)\right)\right)\right], \end{aligned} \tag{4}$$

where $\widetilde{z}$ stands for the noise vector, $G(y, \widetilde{z})$ represents the generator conditioned on a pre-defined labeled y, which differs from the traditional generator [74], [75].

The main focus of this class of methods lies in the design of $\mathcal{L}_{con}$. 3C-GAN [76] is a pioneering work in this area. Regarding the generator, it places significant emphasis on establishing semantic similarity between the generated image $x_v = G(y, \widetilde{z})$ and the corresponding label y. Further, SDDA [77] proposes a consistency loss based on the domain discriminator, which guides the feature extractor to extract domain-invariant features by a binary classification loss. CPGA [78] is inspired by InfoNCE [79] and generates more representative prototypes for each category by adding a contrastive loss.

An alternative approach of virtual domain generation is to simulate the source domain distribution through Gaussian distribution, leveraging the implicit knowledge contained in the source model. Generally, there are mainly two views. One view is to sample noises from standard Gaussian distribution as the inputs of the generator, and the virtual source generation process $\widetilde{x}$ can be expressed as:

$$\widetilde{x} = G\left(\widetilde{z}\right), \widetilde{z} \sim \mathcal{N}(0, 1). \tag{5}$$

Based on data-free knowledge distillation [31], Liu et al. [10] propose a batch normalization statistical loss to model the source domain distribution using the mean and variance stored in the BN layer of the source model.

There are also other methods to regard the source domain distribution as a mixture of multiple Gaussian distributions. For instance, VDM-DA [72] constructs a Gaussian mixture model to represent the distribution of the virtual domain in the feature space as:

$$D^{v}\left(f_{v}\right) = \sum_{k=1}^{K} \pi_{k} \mathcal{N}\left(f_{v} | \mu_{k}, \sigma^{2} I\right), \tag{6}$$

where f_v stands for the virtual features and π_k denotes the mixing coefficients with $\sum_{k=1}^{K} \pi_k = 1$. To estimate the parameters of the Gaussian mixture model, VDM-DA empirically sets K to the number of categories and considers that the weights of the source classifier implicitly contain prototypical information about each category and derives the mean and standard deviation of the model based on the source classifier. SoFA [80] models the inference and generation processes separately, and then generates the mixture of Gaussian distribution by using the predicted classes as a reference distribution.

Utilizing generative models for the virtual domain is not only computationally expensive but also poses a significant challenge to perform domain generalization effectively, particularly when dealing with complex underlying data patterns. Consequently, certain approaches aim to employ a non-generative strategy, such as constructing a virtual source domain by directly selecting reliable data from the target domain. One of the most straightforward approaches is to input the target domain images into the source model, identifying samples with high prediction confidence as closely aligned with the source domain distribution and therefore capable of representing it. However, this method introduces a critical challenge: the issue of limited data for the virtual source domain. For instance, Du et al. [81] discovered that only one-tenth of the target domain's sample size could be selected to construct the virtual source domain, which is inadequate for representing the entire source domain's data distribution.

To address the issue of insufficient data, a range of approaches focus on how to scale up virtual source domain data. Mixup [82] can be regarded as an effective technique and PS [81] mixes the samples in the virtual source domain with each other, which can be expressed as:

$$\begin{aligned} \widetilde{x}_{s,aug} &= \lambda \widetilde{x}_{s}^{i} + (1-\lambda)\, \widetilde{x}_{s}^{j}, \\ \widetilde{y}_{s,aug} &= \lambda \widetilde{y}_{s}^{i} + (1-\lambda)\, \widetilde{y}_{s}^{j}, \end{aligned} \tag{7}$$

where λ denotes the mixup coefficient, $\widetilde{x}_s^i$, $\widetilde{x}_s^j$ are the samples selected by prediction entropy and $\widetilde{y}_s^i, \widetilde{y}_s^j$ are their corresponding labels respectively. In this way, an augmented virtual source domain are obtained by $\widetilde{D}_{aug}^{s} = \widetilde{D}^{s} \cup \widetilde{D}^{aug}$ with $\widetilde{D}^{aug} = \{\widetilde{x}_{s,aug}, \widetilde{y}_{s,aug}\}$. UIDM [83] distinguishes itself by employing a two-step approach. First, it conducts a mixup within the target domain. Second, it takes a comprehensive perspective for construction by considering not only the prediction entropy during the selection of samples, but also incorporating the data uncertainty introduced by the mixup operation and the model uncertainty arising from the dropout operation.

In addition to employing data augmentation through the mixup, another strategy involves the continual transfer of samples from the target domain to the virtual source domain during the training process. Ye et al. [84] propose the utilization of a nonlinear weight entropy minimization loss to progressively reduce the entropy value of high-confidence samples, while leaving low-confidence samples unaffected. This approach results in the generation of more reliable samples. While previous methods have successfully augmented the sample size in the virtual source domain, there remains a notable scarcity of virtual source domain samples for particularly challenging categories. Acknowledging this constraint, ProxyMix [85] addresses this issue by forming a class-balanced proxy source domain through the nearest neighbors of each prototype, coupled with intra-domain mixup. Instead of reconstructing the entire distribution of the source domain, Zhang et al. [86] construct inter-class relationships based on the domain invariance of categories.

2) Intra-Domain Adversarial Alignment: In the preceding section, we have presented the concept of addressing the issue of unavailable source data by generating a virtual domain, reformulating SFDA as a cross-domain problem. In contrast, this category of methods partitions samples in the target domain into two distinct groups based on their confidence levels of model outputs. Subsequently, adversarial learning is employed between the classifier and feature extractor within each group to minimize the quantity of low-confidence samples. The objective of adversarial learning encompasses two aspects: first, high-confidence samples exhibit similarity to the source domain, and the classifier parameters are utilized to preserve the source domain knowledge for such samples; second, low-confidence samples demonstrate domain-specific characteristics in the target domain, and optimizing the feature extractor parameters for these samples enhances the model's generalization capability.

Technically, intra-domain adversarial alignment mostly uses multiple classifiers for adversarial purposes, which can be seen as a variant of bi-classifier paradigm adversarial methods [42], [43], [87] in conventional UDA. Here we first review the generic three-step procedures of the bi-classifier paradigm:

- *Step 1:* Source training:

$$\min_{\theta_{\mathcal{F}}, \theta_{\mathcal{C}_1}, \theta_{\mathcal{C}_2}} \sum_{i=1}^{2} \mathcal{L}_{cls}\left(\mathcal{C}_i\left(\mathcal{F}\left(\mathcal{X}^s\right)\right), \mathcal{Y}^s\right),$$

where $\mathcal{L}_{cls}$ denotes the cross-entropy loss, $\mathcal{C}_1$ and $\mathcal{C}_2$ represent the two classifiers, $\theta_{\mathcal{F}}, \theta_{\mathcal{C}_1}, \theta_{\mathcal{C}_2}$ is the model parameters corresponding to the feature extractor and classifiers respectively.

- *Step 2:* Maximum classifier prediction discrepancy:

$$\min_{\theta_{\mathcal{C}_1}, \theta_{\mathcal{C}_2}} \mathcal{L}_{cls}\left(\mathcal{X}^s, \mathcal{Y}^s\right) - \mathcal{L}_{dis}\left(\mathcal{Y}_1^t | \mathcal{X}^t, \mathcal{Y}_2^t | \mathcal{X}^t\right),$$

where the first term is identical to step 1, $\mathcal{L}_{dis}$ refers to the discrepancy loss of the two classifiers' prediction.

- *Step 3:* Minimize classifier prediction discrepancy:

$$\min_{\theta_{\mathcal{F}}} \mathcal{L}_{dis}\left(\mathcal{Y}_1^t | \mathcal{X}^t, \mathcal{Y}_2^t | \mathcal{X}^t\right).$$

We now discuss this three-step procedure in the SFDA setting. For the first step, the source model is usually trained in the same way. For the second and third steps, the $\mathcal{L}_{cls}$ term is intractable because of the missing source data, and source knowledge can not be well maintained. As a result, the prediction discrepancy term $\mathcal{L}_{dis}$ in this case becomes unanchored.

To solve the above problem in the SFDA setting, a source-specific classifier can be preserved by freezing the parameters, and a target-specific classifier can be trained on the unlabeled target data. Furthermore, the target domain samples can be divided into source-similar and source-dissimilar by the source-specific classifier's prediction confidence or some other criteria, thus the intra-domain adversarial alignment can be achieved on the two data populations. Formally, the intra-domain adversarial alignment methods can be generalized into two steps:

Step 1: Update the target-specific classifier by maximizing the prediction discrepancy between the target-specific classifier and the source-specific classifier on the source-dissimilar samples while minimizing the same prediction discrepancy on the source-similar samples:

$$\min_{\theta_{\mathcal{C}_t}} \sum_{x \in \mathcal{X}_h^t} \mathcal{L}_{dis}\left(p^s(x), p^t(x)\right) - \sum_{x \in \mathcal{X}_l^t} \mathcal{L}_{dis}\left(p^s(x), p^t(x)\right), \tag{8}$$

where $\mathcal{X}_h^t$ and $\mathcal{X}_l^t$ represent the high-confidence (source-similar) and low-confidence (source-dissimilar) samples in the target domain with $\mathcal{X}_h^t \cup \mathcal{X}_l^t = \mathcal{X}^t$, $p^s(x)$ and $p^t(x)$ denote the predictions of source-specific classifier and target-specific classifier, respectively.

Step 2: Update the feature extractor to pull the target features within the two classifiers' decision boundaries:

$$\min_{\theta_{\mathcal{F}}} \sum_{x \in \mathcal{X}^t} \mathcal{L}_{dec}\left(p^s(x), p^t(x)\right), \tag{9}$$

where $\mathcal{L}_{dec}$ stands for the decision loss to improve the generalization ability of the feature extractor.

In general, there are three primary avenues for improving the effectiveness of intra-domain adversarial alignment. These avenues encompass the selection of $\mathcal{L}_{dis}$ and $\mathcal{L}_{dec}$, the partitioning between $\mathcal{X}_h^t$ and $\mathcal{X}_l^t$, and the optimization of network structure. Having outlined the overarching framework for in-domain adversarial alignment, we proceed to discuss the pertinent methodologies within this framework.

A^{2}Net [9], as the first work of intra-domain adversarial alignment, employs a voting strategy to divide the target domain by concatenating the predictions of the source-specific and target-specific classifiers and performing a softmax operation. Specifically, the source-specific classifier's voting score is compared with that of the target-specific classifier, and if the former is higher, the corresponding features are deemed source-similar, and vice versa. Notably, A^{2}Net proposes a Soft-Adversarial mechanism that enables the training of the target-specific classifier and feature extractor through the exchange of voting scores. This mechanism can be mathematically expressed as:

$$\begin{aligned} &\min_{\theta_{\mathcal{C}_t}} -\sum_{i=1}^{n_t}\left(\alpha_i^s \log\left(\sum_{k=1}^{K} p_{(i)k}^{st}\right) + \alpha_i^t \log\left(\sum_{k=K+1}^{2K} p_{(i)k}^{st}\right)\right), \\ &\min_{\theta_{\mathcal{F}}} -\sum_{i=1}^{n_t}\left(\alpha_i^t \log\left(\sum_{k=1}^{K} p_{(i)k}^{st}\right) + \alpha_i^s \log\left(\sum_{k=K+1}^{2K} p_{(i)k}^{st}\right)\right), \end{aligned} \tag{10}$$

where α_i^s, α_i^t represent the voting scores of the source-specific classifier and target-specific classifier respectively, $p^{st} = \sigma([p^s, p^t]^T) \in \mathbb{R}^{2K}$ is the concentration of p^s and p^t after activation. Although the presentation of (10) is different from that of (8), it is still essentially an adversarial training between a feature extractor and a target classifier, so our framework is still applicable.

BAIT [88] is an exemplar approach that conforms to the paradigm of in-domain adversarial alignment. This method utilizes KL divergence as the discrepancy loss and incorporates a bite loss, which serves to enhance the output entropy of the two classifiers. It can be formulated as:

$$\begin{aligned} &\min_{\theta_{\mathcal{C}_t}} \sum_{x \in \mathcal{X}_h^t} \mathcal{D}_{SKL}\left(p^s(x), p^t(x)\right) - \sum_{x \in \mathcal{X}_l^t} \mathcal{D}_{SKL}\left(p^s(x), p^t(x)\right), \\ &\min_{\theta_{\mathcal{F}}} \sum_{i=1}^{n_t} \sum_{k=1}^{K} \left[-p_{i,k}^s \log p_{i,k}^t - p_{i,k}^t \log p_{i,k}^s\right], \end{aligned} \tag{11}$$

where $\mathcal{D}_{SKL}(a, b) = \frac{1}{2}(\mathcal{D}_{KL}(a|b) + \mathcal{D}_{KL}(b|a))$. The KL divergence is commonly used to measure the difference in prediction distributions between two classifiers. However, it does not consider the certainty of classifier outputs, which can lead to ambiguous results. To address this, both D-MCD [89] and DAMC [90] incorporate the CDD distance [91] to evaluate the consistency and certainty of their outputs, while also making changes to their network structures. D-MCD focuses on retaining simple samples early in training by introducing a strong-weak paradigm to filter samples, preventing the inclusion of incorrect high-confidence samples. Alternatively, DAMC uses a network architecture with more than two classifiers to establish a stricter upper bound on the domain gap for classification and determine the optimal number of classifiers based on the number of categories. However, these additional network structures also increase computational overhead.

3) Perturbed Domain Supervision: This class of methods is primarily based on the assumption that both the source and target domain features originate from a domain-invariant feature space [92]. In other words, the source and target domains consist of features that are invariant across domains, while also incorporating some domain-specific factors. From a perturbation perspective, it is feasible to utilize the target domain data to perturb the source data for semantic augmentation in UDA, aiming to guide the model in learning domain-invariant features [93]. Consequently, even in the absence of source domain data, it is also possible to introduce appropriate domain-related perturbations to the unlabeled target data during model training. If the model can effectively withstand these domain-related perturbations, it becomes plausible for the model to acquire domain-invariant features. The overall objective function of perturbed domain supervision can be formulated as follows:

$$\mathcal{L}_{pds} = \mathcal{L}_{cls}\left(\mathcal{F}\left(\mathcal{C}\left(\mathcal{F}\left(\mathcal{X}^{t+}\right)\right)\right), \hat{Y}^{t}\right), \tag{12}$$

where $\mathcal{X}^{t+}$ denotes the perturbed target domain, $\hat{Y}^t$ denotes the pseudo-labels of the model on the original target domain data.

As a representative method in this category, SOAP [94] argues that the ultimate goal of the model is to find domain-invariant features, and expresses the adjustment direction of the model as a vector and reflects it in UDA and SFDA as:

$$\begin{cases} \overrightarrow{\epsilon}\left(\mathcal{D}^t, \mathcal{D}^I\right)_{UDA} = \overrightarrow{\epsilon}\left(\mathcal{D}^t, \mathcal{D}^s\right) + \overrightarrow{\epsilon}\left(\mathcal{D}^s, \mathcal{D}^I\right), \\ \overrightarrow{\epsilon}\left(\mathcal{D}^t, \mathcal{D}^I\right)_{SFDA} = \overrightarrow{\epsilon}\left(\mathcal{D}^{t+}, \mathcal{D}^t\right), \end{cases} \tag{13}$$

where $\mathcal{D}^I$ denotes the domain-invariant feature space, $\overrightarrow{\epsilon}$ denotes the distance in the feature space, and $\mathcal{D}^{t+}$ is a super target domain constructed by adding a target-specific perturbation $\widetilde{N}_T$ to the target domain samples as $X^i_{T+} = \beta X^i_T + (1-\beta)\widetilde{N}_T$. While the concept of a super target domain is appealing, the construction method employed is relatively simplistic through averaging the target images, which may not adequately capture the characteristics of the target domain. Instead of single-step searching, TPDS [95] gradually characterize domain distribution shift by incrementally adding small perturbations along geodesic paths:

$$\mathrm{D}_w\left(\overrightarrow{\epsilon}_{\theta_{k-1}}, \overrightarrow{\epsilon}_{\theta_k}\right) = \max\{d_0, \ldots, d_c, \ldots d_{C-1}\}, \tag{14}$$

where $\mathrm{D}_w(\overrightarrow{\epsilon}_{\theta_{k-1}}, \overrightarrow{\epsilon}_{\theta_k})$ denotes the optimal search direction for the current step and d denotes the infinite Wasserstein distance for different perturbations.

For the same purpose, FAUST [96] interprets this data perturbation from the perspective of uncertainty [97], which first applies a random augmentation to the target images, and then encourages the feature extractor to extract consistent features over the target images before and after the perturbation by both epistemic uncertainty and aleatoric uncertainty. Unlike perturbing target data, VMP [98] introduces perturbations to the parameters of the model by variational Bayesian inference to maintain the model's discriminative power while performing model adaptation. AAA [22], on the other hand, introduces adversarial perturbations [99] into domain adaptation by designing adversarial examples to attack the model, and the generation process of the adversarial examples can be represented as follows:

$$\begin{aligned} \widetilde{x}_t &= x_t + \frac{G\left(x_t\right)}{\left\|G\left(x_t\right)\right\|_2}\epsilon, \\ &\max_{\theta_G} \mathcal{L}_{cls}\left(\mathcal{C}\left(\mathcal{F}\left(\widetilde{x}_t\right)\right), \hat{y}_t\right), \end{aligned} \tag{15}$$

where G denotes the generator, $||\cdot||$ denotes the Euclid norm, $\epsilon > 0$ is the perturbation magnitude to control that the perturbation will not destroy the samples' original semantic information, and $\hat{y}_t$ is the pseudo label corresponding to $\widetilde{x}_t$. The authors argue that if the model defends against these attacks, the generalization ability of the model can be significantly improved in the process. This defense process can be formulated as:

$$\max_{\theta_{\mathcal{F}}, \theta_{\mathcal{C}}} \mathcal{L}_{cls}\left(\mathcal{C}\left(\mathcal{F}\left(\widetilde{x}_t\right)\right), \hat{y}_t\right). \tag{16}$$

In general, the central focus of this class of methods revolves around the meticulous design of perturbations. The first crucial aspect pertains to determining an appropriate magnitude for the perturbation, wherein the objective is to enhance the model's generalization ability without compromising the original properties of the data or the model itself. The second key consideration entails ensuring that the perturbation adequately reflects the distinctive characteristics of the target domain, thus facilitating effective adaptation to the specific task at hand in the target domain.

B. Image-Based Information Extraction

The information within a single image can be primarily categorized into two main components [100]: content information, which is associated with the image label, and style information, which is typically specific to a particular domain. Neighborhood clustering begins by utilizing the content information of the target images and relies on the observation that the target data exhibits a clear structure and clustering distribution. Therefore, maintaining consistency between neighboring nodes can be achieved by increasing the inter-class distance and decreasing the intra-class distance within the feature space. This ultimately reduces classification difficulty, as misclassifications typically occur near the decision boundary of the model [101]. Image Style Translation starts from the style information of the image, and converts the target image to the source style without changing the content information, so that it can be better recognized by the source classifier.

1) Neighborhood Clustering: The key idea behind Neighborhood Clustering is that although the feature distribution in the target domain can deviate from that in the source domain, they can still form clear clusters in the feature space, and the features in the same clusters will come from the same category. Therefore, if the consistency between neighbor nodes in the feature space is encouraged, it enables feature points from the same clusters to move jointly toward a common category. This process can be generally formulated as:

$$\mathcal{L}_{nc} = \sum_{x_j \in \mathcal{N}(x_i)} \mathcal{L}_{dis}\left(p\left(x_i\right), p\left(x_j\right)\right) - \sum_{x_j \notin \mathcal{N}(x_i)} \mathcal{L}_{dis}\left(p\left(x_i\right), p\left(x_j\right)\right), \tag{17}$$

where $\mathcal{N}(x_i)$ represents the neighboring nodes of target sample x_i. G-SFDA [102] is a pioneer job in this category, which proposes to use Local Structure Clustering (LSC) for consistency constraints. And the methodology can be represented as follows:

$$\begin{aligned} \mathcal{L}_{LSC} &= -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}\log\left[p\left(x_i\right)\cdot\mathcal{B}_S\left(\mathcal{N}_k\right)\right] + \sum_{c=1}^{C}\mathrm{KL}\left(\overline{\mathrm{p}}_{\mathrm{c}}||\mathrm{q}_{\mathrm{c}}\right), \\ \mathcal{N}_{\{1,\ldots,K\}} &= \{\mathcal{B}_F^j | top{-}K\left(cos\left(\mathcal{F}\left(x_i\right), \mathcal{B}_F^j\right), \forall \mathcal{B}_F^j \in \mathcal{B}_F\right)\}, \\ \overline{p} &= \frac{1}{n}\sum_{i=1}^{n} p_c\left(x_i\right), \text{and } \mathrm{q}_{\{\mathrm{c}=1,\ldots,\mathrm{C}\}} = \frac{1}{\mathrm{C}}. \end{aligned} \tag{18}$$

From (18), LSC first finds the k-nearest neighbors $\mathcal{N}_k$ to x_i in the feature bank $\mathcal{B}_F$ by calculating the cosine similarity, and then constrains the prediction consistency of the prediction scores of these K features in the scores bank $\mathcal{B}_S$ with x_i. The second term of $\mathcal{L}_{LSC}$ is used to prevent the degenerated solution [103], [104] to encourage prediction balance where p_c is the empirical label distribution and q_c is a uniform distribution. CPGA [78] simplifies this step by constraining only the normalized similarity between neighbor nodes as an auxiliary loss to improve accuracy.

Based on this, NRC [105] further refines the neighbor nodes into reciprocal and non-reciprocal neighbors based on whether they are the nearest neighbors to each other, i.e., $(j \in \mathcal{N}_K^i) \cap (i \in \mathcal{N}_K^j)$, and builds an expanded neighborhood to aggregate more information in the common structure, thereby weighting the affinity of different neighbors as:

$$\mathcal{L}_{\mathcal{N}} = -\frac{1}{n_t} \sum_i \sum_{k \in \mathcal{N}_K^i} A_{ik} \mathcal{B}_{S,k}^T p_i,$$
$$\mathcal{L}_E = -\frac{1}{n_t} \sum_i \sum_{k \in \mathcal{N}_K^i} \sum_{m \in E_M^k} r \mathcal{B}_{S,m}^T p_i, \tag{19}$$

where $A_{ik} = 1$ while i and k are reciprocal nodes, otherwise $A_{ik} = r = 0.1$. Similarly, Tang et al. [106] construct semantic neighbors on the manifold to portray more complete geometric information, Tian et al. [107] combine with pseudo labeling techniques to obtain structure-preserved pseudo-label by the weighted average predictions of neighboring nodes. However, previous methods have only considered maintaining the consistency of the same clusters (reducing the intra-class distance), but ignored the dissimilarity of different clusters (increasing the inter-class distance), AaD [108] achieves this by defining two likelihood function:

$$P(\mathbb{C}_i) = \prod_{j \in \mathbb{C}_i} p_{ij} = \prod_{j \in \mathbb{C}_i} \frac{e^{p_i^T p_j}}{\sum_{k=1}^{N_t} e^{p_i^T p_k}},$$
$$P(\mathbb{B}_i) = \prod_{j \in \mathbb{B}_i} p_{ij} = \prod_{j \in \mathbb{B}_i} \frac{e^{p_i^T p_j}}{\sum_{k=1}^{N_t} e^{p_i^T p_k}}, \tag{20}$$

where neighbor set $\mathbb{C}_i$ includes K-nearest neighbors of node i, and background set $\mathbb{B}_i$ includes the nodes which are not the neighbor of node i. Once these two likelihood functions are obtained, constraints can be placed on both intra-cluster and inter-cluster by the negative log-likelihood as $L_i(\mathbb{C}_i, \mathbb{B}_i) = -\log \frac{P(\mathbb{C}_i)}{P(\mathbb{B}_i)}$.

Neighborhood Clustering is widely used in active SFDA, involving selectively labeling a small portion of the data. This approach offers two main benefits: first, it identifies challenging data points by constructing a neighbor graph. For example, ELPT [109] selects ungraspable data based on KNN, while MHPL [66] considers specific data properties as the most effective choice. Second, label propagation [110], [111] among neighboring nodes can improve the discriminability of the target domain. Essentially, Neighborhood Clustering methods extract structural information from unlabeled target data.

2) Image Style Translation: The goal of Image Style Translation [112], [113] is to transform the same content into various styles, typically accomplished through the use of two loss functions. The first loss function, known as content loss, is employed to ensure consistency in semantic information before and after translation. The second loss function, transfer loss, leverages feature statistics from multiple intermediate layers to achieve style translation. However, prior methods have mostly utilized Gram matrix [100] or instance normalization [114] to incorporate feature statistics, both of which require access to source domain data, rendering them unsuitable for the SFDA setting. To address this problem, Hou et al. proposed a source-free approach by utilizing the mean μ_{stored}^n and variance σ_{stored}^n of the image batches for the nth layer stored in the Batch Normalization (BN) layers [115] as a style representative of the source domain. This process can be formulated as:

$$\mathcal{L}_{ist} = \mathcal{L}_{content} + \mathcal{L}_{style},$$
$$\mathcal{L}_{content} = ||\mathcal{F}^N(\widetilde{x}_t) - \mathcal{F}^N(x_t)||_2,$$
$$\mathcal{L}_{style} = \frac{1}{N} \sum_{n=1}^{N} ||\mu_{current}^n - \mu_{stored}^n||_2 + ||\sigma_{current}^n - \sigma_{stored}^n||_2, \tag{21}$$

where the feature maps in the source classifier have N layers, $\widetilde{x}_t$ denotes the source-styled image through the generator.

CPSS [116], on the other hand, focuses on improving the robustness of the model by augmenting the samples with diverse styles based on AdaIN [114] during training. A model that is insensitive to changes in image style can be obtained and thus has a strong generalization capability. The training loss can be expressed as:

$$\mathcal{L}_{intra} = \sigma\left(\widetilde{F}_{i,j}\right)\left(\frac{F_{i,j} - \mu(F_{i,j})}{\sigma(F_{i,j})}\right) + \mu\left(\widetilde{F}_{i,j}\right),$$
$$\mathcal{L}_{inter} = \sigma\left(\widetilde{F}_{k,i,j}\right)\left(\frac{F_{k,i,j} - \mu(F_{k,i,j})}{\sigma(F_{k,i,j})}\right) + \mu\left(\widetilde{F}_{k,i,j}\right), \tag{22}$$

where $\mathcal{L}_{intra}$ and $\mathcal{L}_{inter}$ represent the intra-image style swap and inter-image style swap among k images. $\widetilde{F}_{i,j}$ and $\widetilde{F}_{k,i,j}$ denote the shuffled patch that provides the style feature and one image can be divided into $n_h \times n_w$ patches.

Similar to CPSS, SI-SFDA [117] simulates learning representations from corrupted images in medical image segmentation by randomly masking some patches with a black background. Improving the utilization of the BN layer is also a concern for this class of methods. Ishii et al. [118] address this issue implicitly by aligning the statistics in the BN layer with the distribution of the target domain, effectively implementing style translation.

V. Model-Based Methods

Model-based methods for SFDA differ from data-based methods, which primarily focus on data generation or exploring data properties. Instead, model-based methods divide the model into multiple sub-modules and adjust the parameters of some of them to adapt to the target domain. One approach to guide the model is self-training, which draws inspiration from semi-supervised learning and leverages the model's outputs on the target domain. In contrast, the self-attention mechanism primarily serves as an auxiliary module owing to its crucial role in facilitating domain adaptation for target localization.

A. Self-Training

Due to the unavailability of supervision for the source data, self-training, also known as self-supervised learning, utilizes the model's predictions on the unlabeled target domain to iteratively refine itself in a self-supervised manner. Within the framework of SFDA, self-training methods predominantly employ pseudo-labeling, entropy minimization, and contrastive learning strategies.

1) Pseudo Labeling: Without supervision, the most intuitive idea is to pseudo-label the target samples based on the predictions of the source model, and then perform self-supervised learning based on these pseudo-labels. Specifically, this process can be divided into three steps: the first step **Prototype Generation** is to generate class prototypes based on the features of some high-confidence samples or all samples, the second step **Peseudo-label Assignment** is to specify pseudo-labels by the distance or similarity between target samples and class prototypes, and the third step **Pseudo-label Filtering** finally filters out some noisy labels to keep the accuracy of the labels. It is worth noting that neither the first or third steps in this process necessarily exist. For example, a sample can be directly pseudo-labeled by the most possible class predicted by the source classifier, but we argue that they have a tight relationship with each other. We will expand on each step in more detail below.

Prototype Generation: A most straightforward way is to select class prototypes by self-entropy [119], [120], i.e., to select samples in each class with self-entropy greater than a certain threshold. Ding et al. [85] define the weights of the source classifier as the class prototypes and found that the mean accuracy is higher than employing the entropy criterion. However, the class prototypes selected in this way may not be representative. To tackle this, some methods [109], [121] are inspired by active learning, and consider target samples with higher free energy to be more representative of the target domain distribution which can be used as class prototypes. Another popular approach is to calculate the centroid [6], [122], [123] of each class based on DeepCluster [124], which can be formulated as follows:

$$c_k = \frac{\sum_{x_t \in \mathcal{X}_t} \delta_k\left(\mathcal{F} \circ \mathcal{G}_t\left(x_t\right)\right) \mathcal{G}_t\left(x_t\right)}{\sum_{x_t \in \mathcal{X}_t} \delta_k\left(\mathcal{F}_t\left(x_t\right)\right)}, \tag{23}$$

where c_k denotes the centroid of kth class, δ_k denotes the kth element in the soft-max operation. Some recent works [85], [125] argue that using only one prototype does not fully characterize the class, so multiple prototypes are generated in each class. In general, this step aims to attain a class prototype p_k that can represent the distribution of each category in the target domain.

Pseudo-label Assignment: Once the class prototype is obtained, the pseudo-label of each sample can be determined by its nearest or most similar class prototype:

$$y_{\tilde{t}} = \arg\min_k \mathcal{L}_{pl}\left(\mathcal{G}\left(x_t\right), p_k\right), \tag{24}$$

where $\mathcal{L}_{pl}$ represents the metric function of how the pseudo-label is given. Note that when no class prototype is generated, the equation degenerates to

$$y_{\tilde{t}} = \arg\max_k \mathcal{F} \circ \mathcal{G}\left(x_t\right), \tag{25}$$

where the pseudo-label is assigned directly by the most confident category of the model outputs. On top of this, another common paradigm [126], [127] is to use a teacher-student model, which involves inputting multiple versions of augmented target images into the teacher model, and then generating pseudo-labels based on the predictions of the teacher model. This process can be represented as:

$$y_{\tilde{t}} = \arg\max \frac{1}{L} \sum_{l=1}^{l=L} \hat{h}_t^l = \arg\max \frac{1}{L} \sum_{l=1}^{l=L} f_{\hat{\theta}_t}(\hat{x}_t^l). \tag{26}$$

where $f_{\hat{\theta}_t}$ denotes the teacher model, L is the total number of augmentations and $\hat{x}_t^l$ is the augmented target image. Moreover, the way of pseudo-labeling is also worth discussing. In the seminal work SHOT [6], it can be observed that the task of assigning pseudo-labels is a time-intensive process, as it necessitates processing all the available data. To tackle this problem, BMD [123] proposes a dynamic pseudo-labeling strategy to update the pseudo-label in the domain adaptation process. Shen et al. [128] label only a subset of the target domain to ensure accuracy. In this labeling process, how to maintain the balance between categories is also a major concern. Qu et al. [123] use the idea of multiple instance learning [129] to form a balanced global sampling strategy, while Ding et al. [130] obtain robust pseudo-labels for target data by utilizing spherical k-means clustering. You et al. [131] set category-specific thresholds to keep the number between categories as consistent as possible.

Pseudo-label Filtering: Due to the presence of the domain shift, the outputs of the model inevitably contain noise, i.e., incorrect pseudo-labels. Therefore, the false label filtering [89], [119], [132] and the noisy label learning [125], [133], [134] are two important directions to improve the accuracy of pseudo labels. False label filtering is mainly to reject unreliable pseudo-labels by designing some reasonable mechanism. One is to design a specific rule mechanism, such as Kim et al. [119] propose an end-to-end filtering mechanism, which only recognizes a sample as reliable when its Hausdorff distance from its most similar class prototype is smaller than its distance from the second similar class prototype; another is to design an optimized network mechanism to perform filtering, such as Chu et al. [89] utilized the advantage that a weak model is more likely to identify hard samples [135], and propose an additional untrained weak network to filter on incorrect pseudo-labels. Yang et al. [132] looked at the stability of negative learning and proposed a multi-class negative learning strategy to learn the filtering threshold of pseudo-label selection adaptively. On the other hand, noise label learning is no longer just filtering wrong pseudo-labels, but also learning useful information from the noisy labels for self-refinement. SHOT++ [134] is an extension of SHOT [6], attempting to improve the reliability of low confidence samples by MixMatch [136] between high confidence samples and low confidence samples for information propagation. NEL [125] introduces ensemble learning [137] into the SFDA setting, first performing data augmentation on the target domain samples with multiple angles based on input and feedback, and then using negative ensemble learning across multiple versions to refine the pseudo label. Recently, Karim et al. [126] applied the idea of curriculum learning to the selection of reliable pseudo labels. After obtaining accurate pseudo-labels through the above steps, a self-supervised loss can be used to guide the model training, typically cross-entropy.

2) Entropy Minimization: Entropy minimization has profound applications in semi-supervised learning [138], [139] and unsupervised learning [140], [141] methods, and was first introduced to the SFDA setting by [6]. Previous unsupervised domain adaptation mostly improves the adaptive capability of the model in the target domain by aligning the feature distributions in the source and target domains, but this is not feasible without the source domain data. Therefore, another idea is to start directly from the results, assuming that a model with adaptive ability has been obtained, then the model outputs of each sample should be deterministic, i.e., entropy minimization, and this ideal result constraint can be inversely employed to guide the optimization of the model. The entropy minimization loss can be expressed as:

$$\mathcal{L}_{ent} = -\mathbb{E}_{x_t \in \mathcal{X}_t} \sum_{k=1}^{K} \delta_k\left(p_t\left(x_t\right)\right) \log \delta_k\left(p_t\left(x_t\right)\right), \tag{27}$$

where p_t denotes the model's prediction concerning sample x_t. Since entropy minimization loss is easy to implement and can be easily combined with other methods such as pseudo labeling methods as a loss function, it has been rapidly adopted for image classification [142], [143], image segmentation [11], [144], medical image analysis [145] and blind image quality assessment [146] in the SFDA setting. However, some works [147], [148] have found that entropy minimization loss may lead to a trivial solution during training, i.e., all predictions are biased towards a certain class, so the batch entropy maximization loss [6] is proposed to ensure the diversity of predictions:

$$\mathcal{L}_{div} = \sum_{k=1}^{K} p_k \log p_k, \tag{28}$$

where $p_k = \mathbb{E}_{x_t \in \mathcal{X}_t}[\delta(p_t^k(x_t))]$, is to accumulate the outputs by class first in one batch, and then perform the soft-max operation. The reason for this is that within a batch, the number of samples in each category should be balanced, so that the uniform distribution has the maximum entropy value. However, this assumption does not hold true in the case of imbalanced class distributions. As such, Tang et al. [149] proposed to assign larger adaptive weights to those classes with a smaller number of samples. CoNMix [150] attempted to use the nuclear-norm maximization (NM) to improves both class-discriminability as well as prediction diversity in a unified way. In many SFDA methods, (27) and (28) are combined to form the information maximization loss:

$$\mathcal{L}_{IM} = \mathcal{L}_{ent} + \mathcal{L}_{div}. \tag{29}$$

Based on the $\mathcal{L}_{IM}$, Ahmed et al. [151] weight the different source domains in the multi-source domain setting to get the weighted information maximization, and Mao et al. [143] incorporate the neighbor node information to graph adaptation. Pei et al. [152] first calculate the transferability of features with uncertainty before entropy minimization, in order to discard irrelevant knowledge from the source domain. However, aside from information maximization, there is currently a relative scarcity of innovations based on entropy minimization. Instead, many approaches simply utilize it as an auxiliary loss to enhance accuracy. The main objective of entropy minimization lies in assessing the outcomes of the model and determining how to restrict its outputs in the target domain, considering various levels such as individual and collective, deterministic and diverse, among others.

3) Contrastive Learning: Self-supervised contrastive learning [153], [154] is a common route used for unsupervised representation learning, which focuses on learning discriminative representations by increasing the distance between negative pairs while gathering positive pairs. This approach is characterized by the construction of positive and negative sample pairs regarding the same sample, which can be mainly categorized as the memory-bank based [155], encoder based [153] and mini-batch based [156], [157]. One most typical contrastive loss is the InfoNCE loss [79], [153]:

$$\mathcal{L}_{cl} = -\sum_{v^+ \in V^+} \log \frac{\exp\left(u^T v^+/\tau\right)}{\exp\left(u^T v^+/\tau\right) + \sum_{v^- \in V^-} \exp\left(u^T v^-/\tau\right)}, \tag{30}$$

where V^+, V^- are the sets of positive and negative pairs about the same sample u respectively, and τ is the temperature parameter. In unsupervised domain adaptation, positive and negative sample pairs are typically constructed using the target domain sample as the key value and the source domain sample as the query. However, in the source-free setting, the latter is unavailable, making it challenging to create representative queries for the source domain. This problem can be primarily addressed through three main approaches. The first manner is to use the source classifier's weights as prototype features for each class of source samples [122]:

$$\mathcal{L}_{cdc}^1 = -\sum_{m=1}^{M} \mathbb{I}_{\hat{y}_u = m} \log \frac{\exp\left(u^T w_s^m/\tau\right)}{\sum_{j=1}^{M} \exp\left(u^T w_s^j/\tau\right)}, \tag{31}$$

where the classifier weight $W_s = [w_s^1, \ldots, w_s^M]$, $\mathbb{I}_{\hat{y}_t{}^i = m}$ is the pseudo label indicator matrix on target sample u. Similarly, [78] uses the generated prototypes for contrastive learning and introduces weight loss to enhance the reliability of the pseudo label. The second way is to increase the similarity of positive pairs in the current batch for contrastive matching [9]:

$$\mathcal{L}_{cdc}^2 = -\log \frac{\exp\left(s_{ij}\right)}{\sum_{v=1}^{b} \mathbb{I}_{v \neq i} |\gamma_{iv}| \exp\left(s_{iv}\right)}, \tag{32}$$

where $s_{ij} = [\sigma(p_i)]^T \sigma(p_j)$ denotes the similarity of the ith sample and the jth sample with the soft-max operation on their model outputs p_i, p_j. b denotes the batch size, γ_{iv} is a threshold function used to determine whether the two samples belong to one class. The last class of approaches preserves a memory bank to store the outputs of the historical models, thus constructing positive and negative sample pairs [158], [159]. HCID [158] uses the current model to encode the samples of the current batch as the query $q^t = M^t(x_q)$, and then uses the historical model to encode the previously stored samples as keys

$k_n^{t-m} = M^{t-m}(x_{k_n})$:

$$\mathcal{L}_{cdc}^{3} = -\sum_{x_q \in X_t} \log \frac{\exp\left(q^T k_+^{t-m}/\tau\right)\gamma_+^{t-m}}{\sum_{i=0}^{N}\exp\left(q^t k_i^{t-m}/\tau\right)\gamma_i^{t-m}}, \quad (33)$$

where N is the number of stored keys, γ is the parameter used to measure the reliability of the keys, and one implementation is the classification entropy. Some work [160] has also employed graph convolutional networks to characterize relations between instances in object detection, thereby guiding the contrastive representation learning. In general, part of the processing of this class of source-free methods is similar to the basic idea of data-based SFDA, i.e., using generative or non-generative approaches to compensate for missing source data.

B. Self-Attention

In computer vision tasks, localizing object regions plays an important role in domain adaptation task [161], but traditional CNN models prefer to capture local domain-specific information, such as background information, which may not help focus objects we care about. Therefore, the self-attention mechanism [162], [163] is introduced into the SFDA setting:

$$\mathcal{L}_{Attn_{self}}(\mathcal{Q},\mathcal{K},\mathcal{V}) = softmax\left(\frac{\mathcal{Q}\mathcal{K}^T}{\sqrt{d_k}}\right)\mathcal{V}, \quad (34)$$

where $\mathcal{Q}$, $\mathcal{K}$, $\mathcal{V}$ denote query, key and value respectively, d_k is the dimensions of $\mathcal{K}$. To alleviate the context loss problem in the source domain, some methods [109], [144], [161] equip the self-attentive module directly with the traditional feature extractor. Equation (34) can also be extended with cross-domain attention [164]:

$$\mathcal{L}_{Attn_{cross}}(\mathcal{Q}_s,\mathcal{K}_t,\mathcal{V}_t) = softmax\left(\frac{\mathcal{Q}_s\mathcal{K}_t^T}{\sqrt{d_k}}\right)\mathcal{V}_t, \quad (35)$$

where $\mathcal{Q}_s$ is from the source domain, $\mathcal{K}_t$ and $\mathcal{V}_t$ are from the target domain, and together they form a sample pair. Thus the loss of self-attention can be roughly formulated as:

$$\mathcal{L}_{sa} = \mathcal{L}_{Attn_{self}} + \mathcal{L}_{Attn_{cross}}, \quad (36)$$

Based on this, CADX [165] divides the target domain image into supported images and query images under the source-free setting, and improves the original patch-to-patch operation to image-to-image in order to capture the overall representations and reduce the computational burden. In addition, some approaches [10], [166] further process the features from both spatial attention and channel attention perspectives to enrich the contextual semantics of the representations. Transformers have recently excelled in various visual tasks [167], [168], incorporating a multi-head self-attention mechanism to process image patches and offer global context. Some approaches have sought to integrate vision transformer (ViT) into SFDA. TransDA [161] attempts to inject Transformer block into a convolutional network as an attention module, and DSiT-B [169] is the first work entirely based on ViT, aiming to enhance domain specificity.

VI. Comparison

Given that the classification task plays a central role in current SFDA methods, we proceed to perform a comparative analysis of the leading SFDA methods on three widely-used classification datasets. In this section, we aim to evaluate the effectiveness of the various modules by modularizing these methods based on the approaches outlined in Sections IV and V. This approach allows us to assess the individual contributions of different modules and gain insights into their impact on overall performance.

A. Comparison Datasets

Office-31 [175] is the dominant benchmark dataset in visual transfer learning, which contains 4,652 images of 31 types of target objects commonly found in office environments, such as laptops, filing cabinets, keyboards, etc. These images are mainly derived from Amazon (online e-commerce images), Webcam (low-resolution images taken by webcam), and DSLR (high-resolution images taken by DSLR). **Office-Home** [176] is a baseline dataset for domain adaptation that contains 4 domains, each consisting of 65 categories. The four domains are Art (artistic images in the form of drawings, paintings, decorations, etc.), Clipart (a collection of clipart images), Products (images of objects without backgrounds, and Real World (images of objects taken with a regular camera). It contains 15,500 images, with an average of about 70 images and a maximum of 99 images in each category.

VisDA-C [177] is a large-scale dataset for domain adaptation from simulators to realistic environments. It consists of 12 common classes shared by three public datasets: Caltech-256 (C), ImageNet ILSVRC2012 (I), and PASCALVOC2012 (P). The source domain samples and target domain samples are synthetic images generated by rendering 3D models and real images, respectively.

To better represent the modules included in each method, we categorize the SFDA methods into four classes: Domain-based Reconstruction (DR), Image-based Information Extraction (IIE), Self-Training (ST), and Self-Attention (SA). More finely, we then divide these four types of methods into nine modules, which are Virtual Domain Generation ($\mathbf{M}_{vdg}$) corresponding to (2), Intra-domain Adversarial Alignment ($\mathbf{M}_{iaa}$) corresponding to (8)–(9), Perturbed Domain Supervision ($\mathbf{M}_{pds}$) corresponding to (12), Neighborhood Clustering ($\mathbf{M}_{nc}$) corresponding to (17), Image Style Translation ($\mathbf{M}_{ist}$) corresponding to (21), Pseudo Labeling ($\mathbf{M}_{pl}$) corresponding to (24), Entropy Minimization ($\mathbf{M}_{ent}$) corresponding to (29), Contrastive Learning ($\mathbf{M}_{cl}$) corresponding to (30) and Self-Attention ($\mathbf{M}_{sa}$) corresponding to (36).

B. Comparison Results

We summarize the classification accuracy results on the Office, Office-Home, and VisDA-C datasets in Tables III, IV, and V. It is important to note that all methods are uniformly set to the single-source homogeneous closed-set source-free domain adaptation and we cite the best results presented in the original paper. In each table, we present the accuracy of the

TABLE III
CLASSIFICATION ACCURACY (%) COMPARISON FOR SOURCE-FREE DOMAIN ADAPTATION METHODS ON THE OFFICE-31 DATASET
(DEFAULT RESNET-50,* FOR VIT)

Method (Source→Target)	DR	IIE	ST	SA	Modules	A → W	D → W	W → D	A → D	D → A	W → A	Avg.
Source-only [170]	-	-	-	-	-	68.4	96.7	99.3	68.9	62.5	60.7	76.1
Target-supervised [134]	-	-	-	-	-	98.7	98.7	98.0	98.0	98.7	86.0	94.3
SHOT [6]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	90.1	98.4	99.9	94.0	74.7	74.3	88.6
3C-GAN [76]	✓				$\mathbf{M}_{iaa}$	93.7	98.5	99.8	92.7	75.3	77.8	89.6
BAIT [88]	✓				$\mathbf{M}_{iaa}$	94.6	98.1	**100.0**	92.0	74.6	75.2	89.4
SHOT++ [134]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	90.4	98.7	99.9	94.3	76.2	75.8	89.2
Kim et al. [119]			✓		$\mathbf{M}_{pl}$	91.1	98.2	99.5	92.2	71.0	71.2	87.2
A[2]Net [9]	✓		✓		$\mathbf{M}_{iaa} + \mathbf{M}_{cl}$	94.0	99.2	**100.0**	94.5	76.7	76.1	90.1
NRC [105]		✓			$\mathbf{M}_{nc}$	90.8	99.0	**100.0**	96.0	75.3	75.0	89.4
ASL [171]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	94.1	98.4	99.8	93.4	76.9	75.0	89.5
TransDA* [161]			✓	✓	$\mathbf{M}_{pl} + \mathbf{M}_{sa}$	95.0	**99.3**	99.6	97.2	73.7	79.3	90.7
CPGA [78]		✓	✓		$\mathbf{M}_{nc} + \mathbf{M}_{pl} + \mathbf{M}_{cl}$	94.1	98.4	99.8	94.4	76.0	76.6	89.9
AAA [22]	✓		✓		$\mathbf{M}_{pds} + \mathbf{M}_{pl} + \mathbf{M}_{cl}$	94.2	98.1	99.8	95.6	75.6	76.0	89.9
VDM-DA [72]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{ent}$	94.1	98.0	**100.0**	93.2	75.8	77.1	89.7
HCL+SHOT [158]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{cl}$	92.5	98.2	**100.0**	94.7	75.9	77.7	89.8
AaD [108]		✓			$\mathbf{M}_{nc}$	92.1	99.1	**100.0**	96.4	75.0	76.5	89.9
U-SFAN [172]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{em}$	92.8	98.0	99.0	94.2	74.6	74.4	88.8
CoWA-JMDS [173]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{pl}$	95.2	98.5	99.8	94.4	76.2	77.6	90.3
CDCL [122]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{cl}$	92.1	98.5	**100.0**	94.4	76.4	74.1	89.3
D-MCD [89]	✓		✓		$\mathbf{M}_{iaa} + \mathbf{M}_{pl}$	93.5	98.8	**100.0**	94.1	76.4	76.4	89.9
BMD + SHOT++ [123]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	94.2	98.0	**100.0**	96.2	76.0	76.0	90.1
SCLM [106]		✓	✓		$\mathbf{M}_{nc} + \mathbf{M}_{pl}$	90.0	98.9	**100.0**	95.8	75.5	76.0	89.4
Jing et al. [98]	✓				$\mathbf{M}_{pds}$	93.3	98.6	**100.0**	96.2	75.4	76.9	90.0
Kundu et al. + SHOT++ [174]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{pl}$	93.2	98.9	**100.0**	94.6	78.3	78.9	90.7
ProxyMix [85]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{pl}$	96.7	98.5	99.8	95.4	75.1	75.4	85.6
UTR [152]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	93.5	99.1	**100.0**	95.0	76.3	78.4	90.3
C-SFDA [126]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{cl}$	93.9	98.8	99.7	96.2	77.3	77.9	90.5
TPDS [95]	✓		✓		$\mathbf{M}_{pds} + \mathbf{M}_{ent}$	94.5	98.7	99.8	97.1	75.7	75.5	90.5
CRS + AaD [86]		✓	✓		$\mathbf{M}_{pl} + \mathbf{M}_{cl} + \mathbf{M}_{nc}$	95.5	99.1	**100.0**	96.6	76.9	78.3	91.1
DSiT-B* [169]			✓	✓	$\mathbf{M}_{ent} + \mathbf{M}_{sa}$	**97.2**	99.1	**100.0**	**98.0**	**81.7**	**81.8**	**93.0**
CoNMix [150]			✓		$\mathbf{M}_{ent} + \mathbf{M}_{pl}$	94.0	98.1	**100.0**	88.8	77.2	75.2	88.9

TABLE IV
CLASSIFICATION ACCURACY (%) COMPARISON FOR SOURCE-FREE DOMAIN ADAPTATION METHODS ON THE OFFICE-HOME DATASET
(DEFAULT RESNET-50, * FOR VIT)

Method	DR	IIE	ST	SA	Modules	Ar → Cl	Ar → Pr	Ar → Rw	Cl → Ar	Cl → Pr	Cl → Rw	Pr → Ar	Pr → Cl	Pr → Rw	Rw → Ar	Rw → Cl	Rw → Pr	Avg.
Source-only [170]	-	-	-	-	-	34.9	50.0	58.0	37.4	41.9	46.2	38.5	31.2	60.4	53.9	41.2	59.9	46.1
Target-supervised [134]	-	-	-	-	-	77.9	91.4	84.4	74.5	91.4	84.4	74.5	77.9	84.4	74.5	77.9	91.4	82.0
SHOT [6]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	57.1	78.1	81.5	68.0	78.2	78.1	67.4	54.9	82.2	73.3	58.8	84.3	71.8
BAIT [88]	✓				$\mathbf{M}_{iaa}$	57.4	77.5	82.4	68.0	77.2	75.1	67.1	55.5	81.9	73.9	59.5	84.2	71.6
SHOT++ [134]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	57.9	79.7	82.5	68.5	79.6	79.3	68.5	57.0	83.0	73.7	60.7	84.9	73.0
A[2]Net [9]	✓		✓		$\mathbf{M}_{iaa} + \mathbf{M}_{cl}$	58.4	79.0	82.4	67.5	79.3	78.9	68.0	56.2	82.9	74.1	60.5	85.0	72.8
NRC [105]		✓			$\mathbf{M}_{nc}$	57.7	80.3	82.0	68.1	79.8	78.6	65.3	56.4	83.0	71.0	58.6	85.6	72.2
TransDA* [161]			✓	✓	$\mathbf{M}_{pl} + \mathbf{M}_{sa}$	67.5	83.3	85.9	74.0	83	84.4	77.0	**68.0**	**87.0**	80.5	**69.9**	**90.0**	79.3
G-SFDA [102]		✓		✓	$\mathbf{M}_{nc} + \mathbf{M}_{sa}$	57.9	78.6	81.0	66.7	77.2	77.2	65.6	56.0	82.2	72.0	57.8	83.4	71.3
CPGA [78]		✓	✓		$\mathbf{M}_{nc} + \mathbf{M}_{pl} + \mathbf{M}_{cl}$	59.3	78.1	79.8	65.4	75.5	76.4	65.7	58.0	81.0	72.0	64.4	83.3	71.6
AAA [22]	✓		✓		$\mathbf{M}_{pds} + \mathbf{M}_{pl} + \mathbf{M}_{cl}$	56.7	78.3	82.1	66.4	78.5	79.4	67.6	53.5	81.6	74.5	58.4	84.1	71.8
PS [81]	✓				$\mathbf{M}_{vdg}$	57.8	77.3	81.2	68.4	76.9	78.1	67.8	57.3	82.1	75.2	59.1	83.4	72.1
AaD [108]		✓			$\mathbf{M}_{nc}$	59.3	79.3	82.1	68.9	79.8	79.5	67.2	57.4	83.1	72.1	58.5	85.4	72.7
U-SFAN [172]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{ent}$	57.8	77.8	81.6	67.9	77.3	79.2	67.2	54.7	81.2	73.3	60.3	83.9	71.9
CoWA-JMDS [173]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{pl}$	56.9	78.4	81.0	69.1	80.0	79.9	67.7	57.2	82.4	72.8	60.5	84.5	72.5
D-MCD [89]	✓		✓		$\mathbf{M}_{iaa} + \mathbf{M}_{pl}$	59.4	78.9	80.2	67.2	79.3	78.6	65.3	55.6	82.2	73.3	62.8	83.9	72.2
Kundu et al. [178]			✓		$\mathbf{M}_{pl}$	61.0	80.4	82.5	69.1	79.9	79.5	69.1	57.8	82.7	74.5	65.1	86.4	74.0
BMD + SHOT++ [123]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	58.1	79.7	82.6	69.3	81.0	80.7	70.8	57.6	83.6	74.0	60.0	85.9	73.6
SCLM [106]		✓	✓		$\mathbf{M}_{nc} + \mathbf{M}_{pl}$	58.2	80.3	81.5	69.3	79.0	80.7	69.0	56.8	82.7	74.7	60.6	85.0	73.1
Jing et al. [98]	✓				$\mathbf{M}_{pds}$	57.9	77.6	82.5	68.6	79.4	80.6	68.4	55.6	83.1	75.2	59.6	84.7	72.8
Kundu et al. + SHOT++ [174]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{pl}$	61.8	81.2	83.0	68.5	80.6	79.4	67.8	61.5	85.1	73.7	64.1	86.5	74.5
FAUST [96]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	61.4	79.2	79.6	63.3	76.9	75.2	65.3	59.4	79.0	74.7	64.2	86.1	72.0
ProxyMix [85]	✓		✓		$\mathbf{M}_{vdg} + \mathbf{M}_{pl}$	59.3	81.0	81.6	65.8	79.7	78.1	67.0	57.5	82.7	73.1	61.7	85.6	72.8
UTR [152]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	59.8	81.2	83.2	67.2	79.2	80.1	68.4	56.4	83.0	73.7	61.2	85.9	73.2
C-SFDA [126]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{cl}$	60.3	80.2	82.9	69.3	80.1	78.8	67.3	58.1	83.4	73.6	61.3	86.3	73.5
TPDS [95]	✓		✓		$\mathbf{M}_{pds} + \mathbf{M}_{ent}$	59.3	80.3	82.1	70.6	79.4	80.9	69.8	56.8	82.1	74.5	61.2	85.3	73.5
CRS + AaD [86]		✓	✓		$\mathbf{M}_{pl} + \mathbf{M}_{cl} + \mathbf{M}_{nc}$	63.5	82.1	85.0	73.0	82.7	82.4	69.5	62.9	82.6	74.2	65.7	87.3	75.9
DSiT-B* [169]			✓	✓	$\mathbf{M}_{ent} + \mathbf{M}_{sa}$	**69.2**	**83.5**	**87.3**	**80.7**	**86.1**	**86.2**	**77.9**	67.9	86.6	**82.4**	68.3	89.8	**80.5**
Tang et al. [149]			✓		$\mathbf{M}_{pl} + \mathbf{M}_{ent}$	58.6	80.2	82.9	69.8	78.6	79.0	67.8	55.7	82.3	73.6	60.1	84.9	72.8
CoNMix [150]			✓		$\mathbf{M}_{ent} + \mathbf{M}_{pl}$	57.6	77.2	82.2	68.4	78.8	78.3	67.1	54.7	81.5	74.0	60.2	85.3	72.1

source-only model [170] (ResNet-50 on Office, Office-Home, and ResNet-101 on VisDA-C) and the target-supervised model at the top. Subsequently, we divide the tasks based on the years 2020, 2021, 2022 and 2023 using horizontal lines that span from the top to the bottom. Our analysis of the comparison results is presented below.

The volume of research on SFDA is experiencing rapid growth: Since 2020, there has been an explosive increase in the number of papers focusing on SFDA. Across the three datasets, the SFDA methods achieve the highest accuracies of only 1.3% (DSiT-B* [169] on Office), 1.5% (DSiT-B* [169] on Office-Home) lower than the accuracy achieved under target supervision (oracle). Even on VisDA-C, the method of Mattia et al. [127] exceeds supervised learning. To a certain extent, SFDA exhibits a tendency to outperform unsupervised domain adaptation methods, both in terms of the quantity of research and experimental performance. This not only showcases the remarkable potential of SFDA but also emphasizes the necessity of a comprehensive SFDA survey.

Self-training is still the most popular research line at this moment: Most of the SFDA methods involve the strategy of self-training. In addition, Entropy minimization [6], [95], [96],

TABLE V
CLASSIFICATION ACCURACY (%) COMPARISON FOR SOURCE-FREE DOMAIN ADAPTATION METHODS ON THE VISDA-C DATASET (DEFAULT RESNET-101, * FOR VIT)

Method	DR	IIE	ST	SA	Modules	plane	bcycl	bus	car	horse	knife	mcycl	person	plant	sktbrd	train	truck	Avg.
Source-only [170]	-	-	-	-	-	55.1	53.3	61.9	59.1	80.6	17.9	79.7	31.2	81.0	26.5	73.5	8.5	52.4
Target-supervised [134]	-	-	-	-	-	97.0	86.6	84.3	88.7	96.3	94.4	92.0	89.4	95.5	91.8	90.7	68.7	89.6
SHOT [6]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{ent}$	94.3	88.5	80.1	57.3	93.1	94.9	80.7	80.3	91.5	89.1	86.3	58.2	82.9
3C-GAN [76]	✓				$\mathbf{M}_{iaa}$	95.7	78.0	69.0	74.2	94.6	93.0	88.0	87.2	92.2	88.8	85.1	54.3	83.3
BAIT [88]	✓				$\mathbf{M}_{iaa}$	-	-	-	-	-	-	-	-	-	-	-	-	83.0
Hou et al. [36]		✓	✓		$\mathbf{M}_{ist}+\mathbf{M}_{ent}$	-	-	-	-	-	-	-	-	-	-	-	-	63.5
SHOT++ [134]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{ent}$	97.7	88.4	90.2	86.3	**97.9**	**98.6**	92.9	84.1	97.1	92.2	93.6	28.8	87.3
Kim et al. [119]			✓		$\mathbf{M}_{pl}$	86.9	81.7	84.6	63.9	93.1	91.4	86.6	71.9	84.5	58.2	74.5	42.7	76.7
A^{2}Net [9]	✓		✓		$\mathbf{M}_{iaa}+\mathbf{M}_{cl}$	94.0	87.8	85.6	66.8	93.7	95.1	85.8	81.2	91.6	88.2	86.5	56.0	84.3
NRC [105]		✓			$\mathbf{M}_{nc}$	96.8	91.3	82.4	62.4	96.2	95.9	86.1	80.6	94.8	94.1	90.4	68.2	85.9
G-SFDA [102]		✓		✓	$\mathbf{M}_{nc}+\mathbf{M}_{sa}$	96.1	88.3	85.5	74.1	97.1	95.4	89.5	79.4	95.4	92.9	89.1	42.6	85.4
ASL [171]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{ent}$	97.3	85.3	86.9	70.7	96.4	72.8	93.0	80.1	95.5	78.1	87.7	50.3	82.8
TransDA* [161]			✓	✓	$\mathbf{M}_{pl}+\mathbf{M}_{sa}$	97.2	91.1	81.0	57.5	95.3	93.3	82.7	67.2	92.0	91.8	92.5	54.7	83.0
CPGA [78]		✓	✓		$\mathbf{M}_{nc}+\mathbf{M}_{pl}+\mathbf{M}_{cl}$	95.6	89.0	75.4	64.9	91.7	97.5	89.7	83.8	93.9	93.4	87.7	69.0	86.0
AAA [22]	✓		✓		$\mathbf{M}_{pds}+\mathbf{M}_{pl}+\mathbf{M}_{cl}$	94.4	85.9	74.9	60.2	96.0	93.5	87.8	80.8	90.2	92.0	86.6	68.3	84.2
VDM-DA [72]	✓		✓		$\mathbf{M}_{vdg}+\mathbf{M}_{ent}$	96.9	89.1	79.1	66.5	95.7	96.8	85.4	83.3	96.0	86.6	89.5	56.3	85.1
PS [81]	✓				$\mathbf{M}_{vdg}$	95.3	86.2	82.3	61.6	93.3	95.7	86.7	80.4	91.6	90.9	86.0	59.5	84.1
AaD [108]		✓			$\mathbf{M}_{nc}$	97.4	90.5	80.8	76.2	97.3	96.1	89.8	82.9	95.5	93.0	92.0	64.7	88.0
U-SFAN [172]	✓		✓		$\mathbf{M}_{vdg}+\mathbf{M}_{ent}$	-	-	-	-	-	-	-	-	-	-	-	-	82.7
HCL+SHOT [158]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{cl}$	93.3	85.4	80.7	68.5	91.0	88.1	86.0	78.6	86.6	88.8	80.0	74.7	83.5
CoWA-JMDS [173]	✓		✓		$\mathbf{M}_{vdg}+\mathbf{M}_{pl}$	96.2	89.7	83.9	73.8	96.4	97.4	89.3	86.8	94.6	92.1	88.7	53.8	86.9
NEL [125]			✓		$\mathbf{M}_{pl}$	94.5	60.8	92.3	87.3	87.3	93.2	87.6	**91.1**	56.9	83.4	93.7	**86.6**	84.2
CDCL [122]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{cl}$	97.3	90.5	83.2	59.9	96.4	98.4	91.5	85.6	96.0	95.8	92.0	63.8	87.5
D-MCD [89]	✓		✓		$\mathbf{M}_{iaa}+\mathbf{M}_{pl}$	97.0	88.0	90.0	81.5	95.6	98.0	86.2	88.7	94.6	92.7	83.7	53.1	87.5
Kundu et al. [178]			✓		$\mathbf{M}_{pl}$	-	-	-	-	-	-	-	-	-	-	-	-	88.2
BMD + SHOT++ [123]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{ent}$	96.9	87.8	90.1	91.3	97.8	97.8	90.6	84.4	96.9	94.3	90.9	45.9	88.7
SCLM [106]		✓	✓		$\mathbf{M}_{nc}+\mathbf{M}_{pl}$	97.1	90.7	85.6	62.0	97.3	94.6	81.8	84.3	93.6	92.8	88.0	55.9	85.3
Kundu et al. + SHOT++ [174]	✓		✓		$\mathbf{M}_{vdg}+\mathbf{M}_{pl}$	-	-	-	-	-	-	-	-	-	-	-	-	87.8
UTR [152]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{ent}$	98.0	92.9	88.3	78.0	97.8	97.7	91.1	84.7	95.5	91.4	91.2	41.1	87.3
FAUST [96]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{ent}$	96.7	77.6	87.6	73.3	95.5	95.4	92.9	83.6	95.3	89.5	87.7	46.9	85.2
ProxyMix [85]	✓		✓		$\mathbf{M}_{vdg}+\mathbf{M}_{pl}$	95.4	81.7	87.2	79.9	95.6	96.8	92.1	85.1	93.4	90.3	89.1	42.2	85.7
C-SFDA [126]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{cl}$	97.6	88.8	86.1	72.2	97.2	94.4	92.1	84.7	93.0	90.7	93.1	63.5	87.8
TPDS [95]	✓		✓		$\mathbf{M}_{pds}+\mathbf{M}_{ent}$	97.6	91.5	89.7	83.4	97.5	96.3	92.2	82.4	96.0	94.1	90.9	40.4	87.6
CRS + AaD [86]		✓	✓		$\mathbf{M}_{pl}+\mathbf{M}_{cl}+\mathbf{M}_{nc}$	**98.1**	90.3	87.2	88.7	97.6	96.3	**93.7**	84.5	**97.5**	**95.3**	91.3	55.0	89.6
DSiT-B* [169]			✓	✓	$\mathbf{M}_{ent}+\mathbf{M}_{sa}$	-	-	-	-	-	-	-	-	-	-	-	-	87.6
Tang et al. [149]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{ent}$	97.4	91.8	87.6	78.1	96.6	99.3	90.6	87.2	95.6	94.6	88.9	57.3	88.7
Mattia et al. [127]			✓		$\mathbf{M}_{pl}+\mathbf{M}_{cl}$	97.3	**96.2**	**90.5**	**91.8**	90.0	94.2	87.4	87.7	97.0	84.3	**93.0**	81.0	**90.0**

[123], [134], [149], [172] as an auxiliary loss can be easily combined with most SFDA methods to improve the discriminability of the target domain representation. Furthermore, contrastive learning [9], [22], [78], [86], [126], [127] generally works with the pseudo-labeling module to play the role of domain alignment. In the absence of source data and target labels, self-supervision through pseudo-labeling for the target data is the most common method. Notably, the highest accuracy 90.0% on VisDA-C dataset was achieved by Mattia et al. [127] that use the the pseudo labeling. This result indicates that pseudo-labeling is a versatile and effective technique, and can serve as a strong baseline for further improvement.

Domain-based reconstruction is commonly used in SFDA: The intuition of SFDA methods behind domain-based reconstruction is quite clear: it aims to reconstruct a new source or target domain to replace the missing source domain for supervision. Intra-domain adversarial alignment methods were the first batch methods of SFDA (BAIT [88] and 3C-GAN [76]) and have subsequently undergone continuous enhancements. The attention of virtual domain generation is on the rise, probably because virtual domain generation methods [72], [85], [173], [174] can be closely combined with self-training methods that play the supervised role of the target domain, thus together further improving the effect.

Neighborhood clustering seems to be effective in image-based information extraction: Compared with image style translation methods [36], neighbor clustering is currently more effective. For example, CRS + AaD [86] achieves the second highest classification accuracy of 91.1% on Office-31 and 89.6% on VisDA-C, implying the neighbor clustering strategy could be a a simple but strong baseline.

Self-attention is promising, especially transformer-based SFDA methods: Although the self-attention-based SFDA methods are uncommon in image classification [102] and mostly seen in semantic segmentation [10], [144], [166], the transformer-based method DSiT-B* [169] achieves the highest accuracy of 80.5% on the Office-Home dataset and 93.0% on the Office dataset. This suggests that encouraging models to turn their attention to the object region may be quite effective for reducing domain shift. Exploring the potential for a more effective integration of the transformer within the SFDA framework could be an intriguing avenue for further investigation.

VII. APPLICATION

Source-free domain adaptation and unsupervised domain adaptation methods [64], [65], [179] are highly overlapped in their application areas. Currently, most of SFDA methods are applied in the field of computer vision and natural language processing.

A. Computer Vision

Most SFDA methods focus on image classification, which is the fundamental task in computer vision. With the popularity of large-scale datasets like VisDA [177] and DomainNet [180], the level of difficulty associated with adaptation tasks is also escalating. Image classification can also extend to various application scenarios such as real-world image dehazing [181], cross-scene hyperspectral image classification [182], and blind image quality assessment [146]. Semantic segmentation methods [10], [11], [52], [84], [132], [166], [183] have also been widely used, including multi-organ segmentation [145], cross-modal

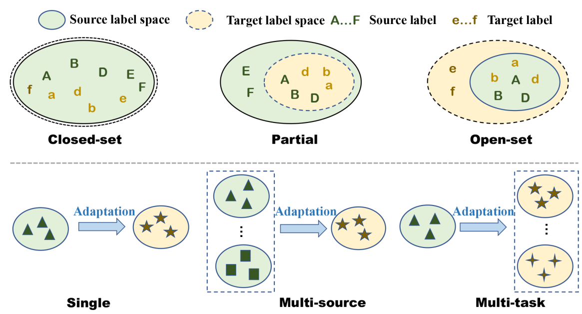

Fig. 3. Illustration of different settings in SFDA.

segmentation [184], [185], cross-device [132], cross-central domain segmentation [117], multi-site and lifespan brain skull stripping [109], RGB-D segmentation [186] and road segmentation [131], [187]. Others applications include object detection [12], [13], [37], [188], person re-identification [189], human pose estimation [190] and video analysis [120], [182] can also benefit from SFDA. Furthermore, many applications of video-based SFDA [191], [192], [193] are also gradually emerging.

B. Natural Language Processing

The application of source-free domain adaption in natural language processing (NLP) is still limited. Related settings and studies in NLP include continuous learning [194], [195] and generalization capabilities of pre-trained models [196]. Laparra et al. designed the SemEval 2021 Task 10 dataset [197] on two tasks, i.e., negation detection and time expression recognition. Su et al. [67] extended self-training [198], active learning [199] and data augmentation [200] to the source-free setting for systematic comparison.

C. Other Related Problems

Domain generalization (DG) [201], [202], which makes the model work on previously unseen target domains, can be seen as a relevant setting for domain adaptation. Notably, emerging source-free DG methods [203], [204] can be considered as a variation of the data-based SFDA approach. Additionally, source-free zero-shot domain adaptation [205] has been proposed to mitigate the need for streaming data in test-time domain adaptation [206]. Furthermore, Ondrej [165] et al. have explored feedforward source-free domain adaption, a back propagation-free method, to enhance user privacy and reduce overhead. Complementary approaches integrate source-free domain adaptation with federated learning [187], black box testing [207], [208], or robust transfer [209] to address diverse practical scenarios.

VIII. Future Research Direction

Despite the substantial advancements in methodology and performance in SFDA, the range of problem settings and research targets investigated in SFDA remains somewhat limited and homogeneous. In the subsequent sections, we delve into potential avenues for expanding the research landscape of SFDA.

A. Enrichment of Weak Research Lines

Although we have discussed numerous SFDA methods in this paper, most of them focus on pseudo-labeling. With the categorization in Fig. 2, some kinds of methods are still less explored, such as perturbed domain supervision, neighborhood clustering, and self-attention, especially transformer-based self-attention mechanisms. Moreover, contrastive learning [210] and image style translation [211] are currently constructed in a relatively uniform and simplistic manner, thereby warranting the introduction of the latest methods [212], [213] within their respective domains. Furthermore, while entropy minimization is extensively employed in SFDA, it primarily serves as an auxiliary loss in the form of entropy loss or information loss, lacking further innovation.

B. Further Theoretical Support

Ben-David et al. [214] derived a general UDA theory that upper-bounds the expected target error based on the distribution divergence of source and target domains, the source error, and the ideal joint error. However, the domain divergence is hard to measure in the source-free setting. Most current SFDA methods are primarily guided by intuition. For instance, Liang et al. [6] proposed information maximization as a means to achieve deterministic and decentralized outputs. Although there have been SFDA theoretical analyses focusing on specific aspects such as pseudo-labeling [89], multi-source domains [151], [215], and model smoothness [216], these analyses are limited to particular methodologies. Therefore, the research on universally applicable theoretical support for SFDA would be highly significant.

C. More Applications

Regarding computer vision tasks, most of the current SFDA methods focus on image analysis, while video data contains significantly more spatial and temporal semantic information, which presents greater processing challenges. Consequently, there are only a few methods specifically designed for video analysis. Furthermore, unsupervised domain adaptation has found numerous applications in natural language processing [217], [218], time-series data analysis [219], [220], recommendation systems [221], [222], and geosciences [223], [224]. However, few of them [225] have been extended to source-free settings, indicating that SFDA may realize its potential in these areas.

D. Extended Settings

This survey mainly covers the single-source closed-set setting of source-free domain adaptation, which is also the most extensively studied case. Nevertheless, as illustrated in Fig. 3 , SFDA can also be extended to partial DA [86], open-set DA [226], universal DA [227], multi-source DA [228], [229], and multi-task DA [150], [230], based on the relationship between the source and target domains. Furthermore, depending on various practical situations, the SFDA setting can be combined with test-time DA [206], [231], federated DA [46], and active DA [232]. In the future, SFDA research is expected to be more task-specific to meet different practical scenarios.

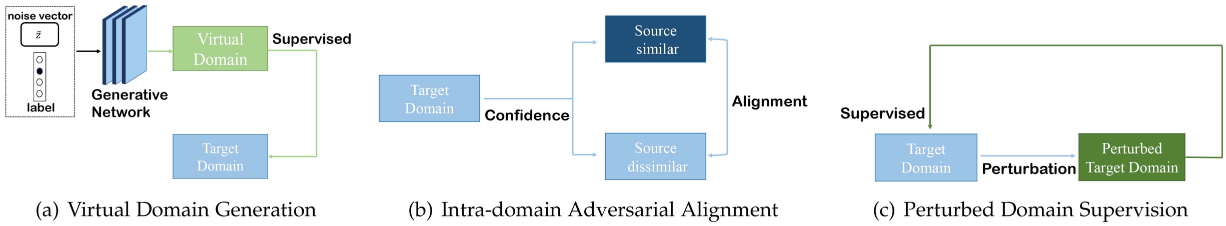

Fig. 4. Overview of Domain Reconstruction methods.

E. Comprehensive Datasets and Evaluation

The majority of datasets currently used for SFDA evaluation have balanced categories and clean data, however, some SFDA research [209] has highlighted the significant impact of unbalanced categories and noise on model accuracy. Additionally, some existing datasets are limited in size and no longer present sufficient challenges to assess the capabilities of current methods. For example, the W to D task on the Office-31 classification dataset [175] has already reached an accuracy of 100%. Thus, more diverse and challenging datasets are necessary for advancing SFDA research. Furthermore, a more comprehensive evaluation scheme such as model robustness and overhead, would provide a more complete model assessment and improve the applicability of SFDA.

IX. Conclusion

Unsupervised domain adaptation is a crucial branch of transfer learning. It aims to transfer knowledge acquired from one labeled domain to another unlabeled domain, reducing the need for extensive annotation. The source-free setting, which does not have access to source domain data, is gaining attention in real-world scenarios due to its privacy and storage efficiency advantages. This paper presents a survey of current source-free adaptation methods and introduces a unified categorization framework. We conduct an in-depth analysis of the experimental results from over 30 representative SFDA methods on three most commonly used datasets Office-31, Office-home, and VisDA. Additionally, we modularize each method to facilitate comprehensive comparisons. To conclude, drawing from our analysis and the present state of SFDA research, we propose potential research directions to advance the SFDA community.

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